

DHEERAJ PAMNANI

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PROFESSIONAL SUMMARY

Analytical professional with over 5 years of experience in data science, AI, and business analytics. Experienced in transforming complex datasets into clear insights that drive decisions across healthcare, education, and finance. Proficient in Python, SQL, Power BI, and cloud platforms, with hands-on work in machine learning, forecasting, and data visualization. Recognized for building high-impact models and dashboards that enable faster, evidence-based strategies.

WORK EXPERIENCE

Research Assistant - AI/ML & Data Science

Dec 2024 – Current

Arizona State University & HonorHealth, Tempe, AZ

- Designing and implementing end-to-end AI/ML pipelines (data ingestion, cleaning, feature engineering, training, evaluation) for mammography datasets (VinDr) using Python, Pandas, NumPy, PyTorch, TensorBoard.
- Led EDA & statistical analysis on large-scale medical imaging data to uncover patterns & ensure data integrity.
- Optimizing object detection workflows with YOLO and RetinaNet (ResNet-101), tuning hyperparameters to improve detection accuracy and robustness.
- Partnered with research teams to interpret complex model outputs, ensuring analytical insights aligned with regulatory requirements and project objectives.

Associate Data Scientist

Mar 2024 – Jul 2024

Yocket, Mumbai, India

- Developed predictive models using Python (Scikit-Learn, TensorFlow) that forecasted student enrolment trends and improved campaign targeting accuracy by 23%.
- Automated ETL pipelines on AWS and Databricks to consolidate data from CRM and Google Analytics, reducing reporting latency by 40%.
- Conducted exploratory data analysis (EDA) on user engagement metrics using SQL and Power BI, uncovering key drivers of lead conversion.
- Collaborated with product and marketing teams to translate model insights into actionable recommendations, driving a 15% increase in student acquisition efficiency.
- Optimized ML models through hyperparameter tuning and feature selection, enhancing overall model precision from 78% to 91%.

Business Data Analyst

Dec 2022 – Feb 2024

Azent Overseas Education Ltd, Mumbai, India

- Designed KPI dashboards in Power BI to track B2B partner performance and cut manual reporting time by 35%.
- Delivered actionable insights that optimized lead prioritization, resulting in an 18% increase in conversion rates and improved partner engagement strategies.
- Streamlined CRM data workflows using SQL and Excel automation, enabling real-time visibility into pipeline progression.
- Partnered with leadership to forecast quarterly revenue through regression and time-series analysis, increasing forecast accuracy by 25%.
- Conducted pricing and market trend analysis to support partnership expansion strategies, resulting in three new international collaborations.

Business Development Analyst

Jan 2021 – Nov 2022

Byju's, Mumbai, India

- Developed interactive Tableau dashboards to visualize regional sales and agent performance metrics, driving a 40% improvement in productivity tracking and decision-making.
- Analyzed over one million sales transactions using SQL and Excel (PivotTables, Power Query) to identify underperforming markets and product trends.
- Developed data-driven sales strategies that increased monthly revenue by 22% through improved territory allocation.
- Performed cohort and churn analysis to measure customer retention, helping optimize re-engagement campaigns.
- Delivered executive reports combining financial modeling and performance KPIs to support data-driven decision-making.

IT Analyst

NCR Corporation, Gurugram, India

Apr 2018 – Jul 2019

- Provided technical analysis and support for POS systems across 50+ client sites, maintaining 99.5% uptime.
- Supported database maintenance and executed data migration tasks using SQL and Java, enhancing data consistency and reliability across multiple retail systems.
- Supported software deployment and testing activities in Agile sprints, reducing defect resolution time by 20%.
- Documented system configurations and workflows in Confluence for audit compliance and knowledge transfer.
- Collaborated with cross-functional teams to enhance data security protocols, strengthening compliance with corporate IT policies.

PROJECTS

Capstone: Financial Forecasting Dashboard (MedAire, Inc.)

Jan 2025 – May 2025

- Developed an interactive Power BI dashboard using DAX to forecast contractual revenue, improving budgeting accuracy & accelerating cross-functional decisions with dynamic KPIs & sensitivity levers.
- Evaluated Time Series & Exponential Smoothing models in Python, reducing RMSE to 5K in revenue predictions.
- Built data pipeline with Python libraries (Pandas, NumPy) to identify trends & improve prediction accuracy.

Modern NLP Applications: BERT for Business Strategy

Feb 2025 – Mar 2025

- Engineered BERTopic model with UMAP on 100K+ Yelp tips (unstructured data) to uncover 30 high-impact customer experience themes, enabling actionable segmentation by sentiment, cuisine, and location.
- Performed EDA with Pandas and Matplotlib to prepare data for BERT modeling, enhancing clustering quality.
- Achieved 92% sentiment classification accuracy with Neural Nets, driving insights by cuisine & location for business strategy.

Real-time Wait Time Estimation at ASU (Computer Vision Project)

Apr 2025 – May 2025

- Built a YOLOv8-based model to count customers, improving eatery queue monitoring accuracy using spatial filtering techniques.
- Trained a regression (MLP) on image-derived counts to estimate wait times with high predictive correlation.

ChatGPT Market Positioning & Segmentation (Python + Enginius)

Feb 2025 - Mar 2025

- Designed and executed a perceptual survey with 78 respondents, applying Principal Component Analysis (PCA) to extract and visualize two key positioning dimensions that explained 99.8% of the variance in customer perceptions.
- Leveraged K-Means clustering to segment users into distinct groups based on loyalty metrics, enabling the recommendation of a targeted pricing strategy of \$20–25 per month for the highest-loyalty customer segments.

SKILLS	
Programming language	: Python, SQL, Java, Jupyter Notebooks
Data Analytics & Visualization	: Power BI, Tableau, Looker, Excel (Advanced), dbt, Databricks
Statistics	: Regression, Time Series Analysis, Hypothesis Testing, ANOVA, Bayesian Methods
Machine Learning & AI	: Scikit-Learn, TensorFlow, PyTorch, NLP (BERT, GPT, LLMs), Deep Learning (CNNs)
Cloud & Databases	: AWS (S3, SageMaker, Redshift), Azure, GCP, Snowflake, BigQuery, Spark
Business & Strategy	: KPI Design, Forecasting, Product Analytics, Financial Modeling, Strategic Planning
Collaboration & Delivery	: Agile/Scrum, Stakeholder Management, Data Storytelling, Communication
Other Tools	: Git/GitHub, ETL Pipelines, Data Warehousing, Data Governance

EDUCATION

Master of Science in Business Analytics | Arizona State University, Tempe, AZ

Aug 2024 – May 2025

Bachelor of Technology in Computer Science & Engineering | Amity University, Noida, INDIA

Jul 2014 – Jun 2018

LEADERSHIP & ACTIVITIES

- Member, Supply Chain Management Association (SCMA), Arizona State University - Led analytics workshops and coordinated events bridging students with industry professionals.
- Research Assistant – Breast Cancer AI, ASU & HonorHealth - Collaborated with cross-disciplinary teams, mentored peers on ML pipeline optimization.
- Presenter, ASU MSBA Capstone Showcase - Delivered poster on Financial Forecasting Dashboard, demonstrating advanced forecasting with Python & Power BI to faculty and industry judges.